1.What is Linux?

Linux is a free, open-source operating system kernel first released by Linus Torvalds in 1991 while he was a student at the University of Helsinki. Technically, the word "Linux" refers only to the kernel itself - the core program that manages a computer's hardware and lets other software communicate with it.

In everyday use, however, "Linux" usually means a complete operating system built around that kernel, bundled together with tools from the GNU Project and other software. These complete packages are called distributions, or "distros" - well-known examples include Ubuntu, Debian, Fedora, and Red Hat Enterprise Linux (RHEL).

Because Linux is open source, its code is publicly available for anyone to inspect, modify, and redistribute under licenses such as the GNU General Public License (GPL). This has allowed a huge global community of developers and companies to continuously improve it. Today Linux quietly powers a large share of the world's technology - most web servers and cloud infrastructure, the majority of supercomputers, Android phones, smart TVs, routers, and countless embedded devices all run on it.

2.Why Do DevOps Engineers Use Linux?

DevOps is built around automating, deploying, and reliably running software - and Linux sits at the center of almost all of it. Main reasons DevOps engineers rely on it:

- Server dominance - most cloud servers and production environments (AWS, Azure, GCP) run Linux, so deploying and troubleshooting infrastructure requires Linux skills.
- Automation-friendly - Linux's command-line shell makes it easy to script repetitive tasks, which is central to CI/CD pipelines and infrastructure automation.
- Free and open source - no per-server licensing fees, which matters enormously at the scale of hundreds or thousands of machines.
- Stability and uptime - Linux servers can run for months or years without a required reboot, which matters for production systems.
- Tooling ecosystem - core DevOps tools such as Docker, Kubernetes, Jenkins, Ansible, and Terraform were built for Linux first and run most efficiently on it.
- Lightweight footprint - Linux can run without a graphical interface, leaving more CPU and memory for the actual workload.
- Fine-grained control - Linux gives engineers deep control over permissions, processes, and networking.

3.Difference Between Windows and Linux
windows
Cost -Paid, proprietary license
Source code- Closed - owned by Microsoft
Customization -Limited customization
Interface -Graphical-first, with PowerShell/CMD available
Installing software - .exe / .msi installers
File system - NTFS - case-insensitive
Typical use -Desktops, gaming, enterprise office use
Security model - More widely targeted due to desktop popularity

Linux:
Cost - Free and open source
Source code- Open - anyone can view or modify it
Customization -Highly customizable, down to the kernel
Interface - Command-line first, optional desktop environments (GNOME, KDE)
Installing software - Package managers (apt, dnf, yum)
File system - ext4, XFS, Btrfs - case-sensitive
Typical use - Servers, cloud, embedded systems, developers
Security model- Strict permissions, smaller attack surface

In short, Windows is generally chosen for familiarity and desktop or office use, while Linux is chosen for
servers and infrastructure where cost, control, and stability matter most - exactly the environment DevOps
engineers work in.

4. What is an Operating System?
An operating system (OS) is the fundamental software that manages a computer's hardware and provides
a platform on which applications can run. It sits between the user, the applications, and the physical
hardware, translating requests from one into instructions the other can understand. Core responsibilities
include:
• Process management - deciding which programs run, when, and for how long.
• Memory management - allocating and freeing RAM as programs need it.
• File system management - organizing and storing data on disks.
• Device management - communicating with hardware such as keyboards, disks, and network cards
through drivers.
• User interface - providing a way, graphical or command-line, for people to interact with the machine.
• Security and access control - managing permissions for different users and processes.
Linux, Windows, macOS, and Android are all examples of operating systems.

5. What is the Linux Kernel?
The kernel is the core of any operating system - the part that directly manages hardware resources like
the CPU, memory, and connected devices, and mediates every request that software makes of the
hardware.
The Linux kernel specifically was created by Linus Torvalds and first released in 1991. It is open source
under the GPL license and has since been developed collaboratively by thousands of contributors and
companies worldwide, including Google, IBM, Intel, and Red Hat. Its core jobs include:
• Process scheduling - deciding which task the CPU works on at any given moment.
• Memory management - allocating RAM to processes safely and efficiently.
• Device drivers - letting software communicate with hardware.
• System calls - the interface applications use to request services from the kernel.
• File system handling - managing how data is read from and written to storage.
Different Linux distributions - Ubuntu, Fedora, Debian, and others - all share this same core kernel, but
package it with different software, defaults, and desktop environments on top.

6. What is Ubuntu?
Ubuntu is one of the most popular Linux distributions, developed and maintained by Canonical Ltd. It is
based on Debian and was first released in 2004 with the goal of making Linux approachable for everyday
users and organizations, not just experienced engineers.
• Free and open source, with a large, active community and strong documentation - a common starting
point for people learning Linux.
• Follows a predictable release cycle: a new version every six months, plus Long Term Support (LTS)
releases every two years that receive five years of security updates.
• Used both as a desktop OS and, very heavily, as a server OS - a large share of cloud instances on
AWS, Azure, and other providers run Ubuntu Server.
• Uses the APT package manager (commands like apt install and apt update) to install and manage
software.
Its ease of use and strong support make Ubuntu a common choice for developers and DevOps engineers
who are learning Linux or standing up new servers - likely why it is the distribution used later in Task 2 of
this course.

7. Why Do Servers Mostly Use Linux?
Linux powers the large majority of the world's servers, cloud platforms, and

enterprise infrastructure. This
comes down to a few practical advantages:
• Cost - Linux is free, a major saving when running thousands of servers compared to per-machine
licensing fees.
• Stability - Linux servers are known for very high uptime and rarely need restarting, even after routine
updates.
• Performance - without a mandatory graphical interface, Linux leaves more CPU and memory
available for the actual workload.
• Security - a strict permission model, a smaller attack surface, and fast open-source patching suit
security-conscious environments.
• Flexibility - administrators can install only what is needed, producing lightweight, purpose-built images
ideal for cloud computing and containers.
• Remote management and automation - Linux is built around the command line and tools like SSH,
making it easy to manage and automate at scale.
• Ecosystem fit - nearly all major cloud-native tools, from Docker and Kubernetes to Nginx, were built
with Linux as the primary target.
Together, these factors are why Linux, not Windows, is the default choice for servers, cloud infrastructure,
and the environments DevOps engineers work in every day.